

Notes: Machine Learning

Title: Machine Learning Study Notes

Introduction to Machine Learning:

- Machine learning is a subset of artificial intelligence that focuses on enabling systems to learn and improve from experience without being explicitly programmed.
- It involves algorithms that allow computers to find patterns and insights in data and make decisions or predictions based on that data.

Types of Machine Learning:

1. Supervised Learning:

- In supervised learning, the algorithm is trained on a labeled dataset, where each data point is associated with the correct output.
- The goal is to learn a mapping from input variables to output labels.
- Examples include regression and classification tasks.

2. Unsupervised Learning:

- Unsupervised learning deals with unlabeled data, where the algorithm explores the data to find hidden patterns or structures.
- The goal is to learn about the inherent structure of the data.
- Examples include clustering and dimensionality reduction.

3. Reinforcement Learning:

- Reinforcement learning is a type of machine learning where an agent interacts with an environment to learn how to make decisions.
- The agent receives rewards or penalties based on its actions, which guides it towards optimal behavior.
- Examples include game playing and robot control.

Key Concepts in Machine Learning:

1. Features and Labels:

- Features are the input variables or attributes of a dataset, while labels are the outputs or predictions.
- The goal of machine learning is to learn the relationship between features and labels to make accurate predictions.

2. Training and Testing Data:

- A dataset is typically divided into training and testing sets to evaluate the performance of a machine learning model.
- The model is trained on the training data to learn patterns, and then tested on the unseen testing data to assess its generalization ability.

3. Model Evaluation:

- There are various metrics and techniques used to evaluate the performance of a machine learning model, such as accuracy, precision, recall, F1 score, and ROC curve analysis.

Popular Machine Learning Algorithms:

1. Linear Regression:

- Linear regression is a supervised learning algorithm used for predicting continuous outcomes.
- It fits a linear relationship between the input features and the target variable.

2. Decision Trees:

- Decision trees are versatile machine learning algorithms that can perform both classification and regression tasks.
- They make decisions by splitting the input space into regions based on feature values.

3. Support Vector Machines (SVM):